DwarfJeans Analysis Pipeline

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Abstract

We present a spherical Jeans modelling pipeline for Milky Way satellite dwarf galaxies. Per-star line-of-sight velocities are ingested via two paths: a monolithic Geha et al. (2026) catalog (Path A) and a collection of per-paper adapters that stage data from individual published tables (Path B), together covering 39 galaxies in a registry-driven schema stored in `data/registry/galaxies.ecsv`. The physical model couples an NFW dark-matter halo (scale radius r_s, scale density ρ_s) with a Plummer stellar tracer and a constant velocity anisotropy β, solved via the projected spherical Jeans equation. Inference proceeds with DYNESTY nested sampling over seven parameters — $\bar{V}$, $\log_{10}\rho_s$, $\log_{10}r_s$, a symmetrized anisotropy $\tilde{\beta}$, and three Gaussian nuisances marginalising over distance, ellipticity, and projected half-light radius — under a user-selectable prior on $(\ln\rho_s, \ln r_s)$: uniform, log-uniform, or conditional Jeffreys. Posterior samples are propagated to $\sigma_{\rm los}(R_{1/2,\rm 2D})$, $M(r_{1/2})$, and tidally-truncated NFW J- and D-factors at four reporting angles ($0.1°$, $0.2°$, $0.5°$, and α_c or $\alpha_c/2$ respectively). Calibration uses a single deterministic Asimov realisation and a 15-realisation Monte Carlo mock-recovery study on synthetic UFD galaxies, following the gold-standard convention of generating synthetic input at known truth and reporting bias and dispersion across realisations.

This document describes the pipeline as implemented in code. Repository layout, module boundaries, and migration history live in `ARCHITECTURE.md` *at the repository root; the working design plan, P&S 2018 deviation log, and per-stage extended treatments (including derivations) live in* `docs/plan/` *(overview entry point:* `pipeline_overview.md`*). Each section below opens with a pointer to the relevant plan document so a reader can drill into the derivation, the convention, or the historical rationale without chasing references through the prose. This writeup does not duplicate either the architecture file or the plan documents.*

Contents

1 Stage 0a: Galaxy registry

The full provenance log for the LVDB pin, the staging conventions, and the per-source rationale lives in `docs/plan/data_sources.md` (LVDB section); this section summarizes the as-implemented build.

1.1 Data source and study sample

The galaxy registry is built by `src/dwarfjeans/ingest/stage0a_registry.py` from the Local Volume Database (LVDB) pinned to version 1.0.5 (Zenodo DOI 10.5281/zenodo.15476348). The combined catalog is staged on disk at `data/lvdb_v1../comb_all.csv`; the build script verifies a SHA-256 checksum against `data/lvdb_v1../checksums.sha256` and aborts on mismatch. No network access occurs at run time.

The 39-galaxy study sample is enumerated in `src/dwarfjeans/ingest/config/study_sample.yaml`. Each entry carries four fields: `lvdb_key` (the LVDB `key` string used to select the row from `comb_all.csv`), `study_name` (the human-readable label used in plots and tables), `path` (`A` for Geha et al. 2026 DEIMOS data or `B` for the LVDB `ref_vlos` paper; used by Stage 0b), and `geha_galaxy` (Path A only: the abbreviation matching the `system_name` column of `data/geha2026/table5A_20260110.csv`). The study sample contains 22 Path A galaxies and 17 Path B galaxies.

After loading `comb_all.csv`, the build script applies three mandatory filters. First, every study-sample `lvdb_key` must resolve to exactly one row in `comb_all.csv`; an unmatched key is a hard error. Second, all 39 rows must satisfy `confirmed_dwarf` = 1. Third, each row must have non-null values for `rhalf`, `distance_modulus`, and `apparent_magnitude_v`. Host-chain resolution verifies that every galaxy traces transitively to the Milky Way; the immediate host is read from the LVDB `host` column and the chain is followed through the full `comb_all.csv` so that subhalo hosts such as `lmc` $\rightarrow$ `mw` resolve correctly. The resolved chain is stored as a `->`-delimited string (e.g. `lmc->mw` for the four Carina-region LMC satellites).

1.2 Coordinate and structural transformations

Distance in kpc is derived from the distance modulus μ as

$$d\,[\mathrm{kpc}] = 10^{(\mu-10)/5}. \tag{1}$$

The LVDB `rhalf` column gives the 2D major-axis half-light radius $R_{1/2,\mathrm{maj}}$ in arcminutes; it is converted to physical units as

$$R_{1/2,\mathrm{maj}}\,[\mathrm{pc}] = \theta\,[\mathrm{arcmin}] \times \frac{\pi}{180\times 60} \times d\,[\mathrm{kpc}] \times 1000. \tag{2}$$

The azimuthally averaged (sphericalised) 2D half-light radius, which is the quantity entering the Jeans likelihood as $r_{1/2}$ in P&S 2018 eq. 8, is

$$r_{1/2}\,[\mathrm{pc}] = R_{1/2,\mathrm{maj}}\sqrt{1-\varepsilon}, \tag{3}$$

where $\varepsilon = 1 - b/a$ is the LVDB ellipticity. The LVDB pre-derived columns `rhalf_physical` and `rhalf_sph_physical` are never consumed as inputs; they serve only as cross-checks against equations (2)–(3) to tolerances of 3% (major-axis) and 7% (sphericalised). The wider sphericalised tolerance reflects Jensen's inequality: the LVDB values are MC medians over the joint ($\texttt{rhalf}, \varepsilon, d$) error distributions, which the concave $\sqrt{1-\varepsilon}$ factor systematically lowers relative to our deterministic central value. Empirically the worst observed offsets are 1.6% for the major-axis check (Tucana V) and 5.5% for the sphericalised check (Leo V); a failure on either check aborts the build.

1.3 Ellipticity missing-data convention

When a galaxy has no measured ellipticity in LVDB v1.0.5 (`ellipticity` null), the registry records `ellipticity_missing=True` and the deterministic computation uses $\varepsilon = 0$ in equation (3); this is the correct central value under the uniform prior $U(0, 0.5)$ that Stage 2 substitutes for the missing measurement. The `ellipticity_ul` column preserves any upper-limit value published by LVDB (e.g. Crater II has `ellipticity_ul` = 0.10) for diagnostics; it does not enter any Stage 0a computation.

1.4 Spatial model and Plummer scale radius

LVDB v1.0.5 does not provide a per-galaxy photometric profile flag. The spatial model is read from a hand-curated override file, `src/dwarfjeans/ingest/config/spatial_model_overrides.yaml`, which specifies a default (currently `plummer`, matching the P&S 2018 convention for ultra-faint dwarfs) and a per-galaxy `overrides` dict; the overrides dict is currently empty so all 39 galaxies use the Plummer model. For `spatial_model="plummer"`, the Plummer scale radius equals the sphericalised half-light radius: $r_p = r_{1/2}$. An override to `"exponential"` would require the conversion $r_p = 1.68\, r_{\rm exp}$ (P&S 2018 §3 footnote 4) and is currently guarded by `NotImplementedError` because LVDB does not expose a per-galaxy $r_{\rm exp}$ value.

1.5 Output schema

The registry is written to `data/registry/galaxies.ecsv` as an Astropy ECSV v1.0 file with units in column metadata. The 39 rows are ordered to match `study_sample.yaml`. A human-readable build log is written in parallel to `data/registry/build_log.txt`. Columns group by purpose as follows; column-by-column provenance is in `docs/plan/data_sources.md`.

- **Identification**: `lvdb_key`, `study_name`, `path` (A/B), `geha_galaxy` (Path A label), `host`, `host_chain`.
- **Sky position**: `ra_deg`, `dec_deg` [deg].
- **Structural inputs**: `rhalf_arcmin` (with ± columns), `ellipticity` (with ± columns and `ellipticity_ul`), `ellipticity_missing` flag, `spatial_model`.
- **Derived radii**: `rhalf_major_pc` (eq. 2), `r_half_2d_pc` (eq. 3), `plummer_radius_pc` ($= r_{1/2}$ under the current default).
- **Distance**: `distance_modulus` (with ± columns), `distance_kpc` (eq. 1).
- **Kinematics**: `vlos_systemic_kms` (with ± columns), `vlos_sigma_kms`, `vlos_sigma_unresolved` flag. The LVDB $\sigma_{\rm los}$ value is metadata only; it does not enter the Jeans likelihood.
- **Photometry**: `M_V`.
- **LVDB provenance**: `ref_vlos`, `ref_structure` — NASA ADS bibcodes for the LVDB historical kinematic and structural references. These are pedigree fields and are *not* guaranteed to identify the catalog actually consumed by our analysis.
- **Analysis source**: `data_source_paper` — bibcode of the paper whose per-star catalog is loaded into `data/star_catalogs/<lvdb_key>.npz`. For Path A this is `Geha2026arXiv260210200G` regardless of LVDB's `ref_vlos`; for Path B this matches `ref_vlos` (the adapter dispatch key). Downstream code, writeup citations, and comparisons against published work should use this column, not `ref_vlos`.

2 Stage 0b: Per-galaxy stellar kinematic ingest

The two-path ingest design (Geha 2026 monolithic catalog vs. per-paper adapters) and the no-cuts-at-ingest invariant are documented in `docs/plan/data_sources.md`; per-paper combiner conventions (IVW recipes, $\sigma_{\rm sys}$ floors, instrument zero-point offsets) are tabulated in `docs/plan/per_paper_combiners.md`. This section summarizes the two paths as they are implemented.

2.1 Path A: Geha+2026 ingest

The 22 Path A galaxies are ingested from the homogeneous Keck/DEIMOS spectroscopic census of Geha et al. (2026, Papers I/II). The driver is src/dwarfjeans/ingest/stage0b_geha.py; it reads the pre-staged full-precision per-star catalog data/geha2026/table5A_20260110.csv (Paper I, 24,436 rows × 50 columns), filters per-galaxy slices by the system_name column (matched to the registry field geha_galaxy), and writes one data/star_catalogs/<lvdb_key>.npz archive per galaxy. Table 5A is the authoritative source as of 2026-05-08 because it carries the binary Pmem_novar column (1 = member with velocity variables already removed; 0 = non-member or variable) used by Paper II §3.1 to compute the Table A1 N counts. The legacy table3A_20260110.csv release exposes only a graded Pmem and a separate Var flag; combining them as $\mathrm{Pmem} > 0.5\,\&\,\mathrm{Var} \neq 1$ does not exactly reproduce Paper II's N.

Output schema. Every Path A archive carries the canonical columns required by the downstream Jeans inference: V and sigma_eps (heliocentric line-of-sight velocity and 1σ error in $\mathrm{km\,s^{-1}}$, mapped from v and v_err); p (binary 0/1 membership flag, copied verbatim from Pmem_novar); R (projected physical radius in kpc, computed via staging.projected_radius_kpc); star_id (zero-based contiguous integer per galaxy, with per-star uniqueness verified); and RA_star, Dec_star. Auxiliary columns retained verbatim include FeH, FeH_err, Var (from flag_var), Pmem (the graded probability, kept for cross-checks), g_mag, r_mag, MV, nmask, t_exp, SN, CaT, CaTerr. Stage 0b is raw-data-only: membership thresholding and aperture cuts are deferred to a downstream selection stage.

2.2 Path B: per-paper adapters

The remaining 17 galaxies are ingested from 15 individual publications via the per-paper adapter system in src/dwarfjeans/ingest/stage0b_pathb.py dispatching to modules under src/dwarfjeans/ingest/path_b_adapters/.

Adapter contract. Each adapter exposes a single function

```
load(staged_dir,registry_row)->(arrays,meta_extra)
```

where staged_dir points to the checksum-verified raw-data directory and registry_row carries the LVDB metadata for the target galaxy. The driver requires six array keys (V, sigma_eps, p, star_id, RA_star, Dec_star), all of identical length, plus a meta_extra dict that is merged into the archive's _meta JSON blob. Adapters with catalog_granularity="per_star" must emit unique star_id values; per-epoch adapters are permitted to repeat a star_id across rows (one per observation). Dispatch is by bibkey <lastname><year> derived from the LVDB ref_vlos field; unknown adapters and unknown instrument tags raise rather than falling back silently.

Papers and galaxies covered. Table 1 lists the 15 adapter modules (covering 17 galaxies) currently in the Path B ingest.

Per-star adapters emit pre-combined velocities directly; per-epoch adapters emit one row per spectrograph exposure, which the multi-epoch combiner (Section 2.3) reduces to per-star rows. Several per-epoch sources include stars flagged as binary candidates by the source authors (e.g. Tuc V-1 in Hansen+2024, TucII-309 / TucII-078 in Chiti+2023); these flags are carried as auxiliary columns but no rows are dropped at ingest. Binary-aware sample selection is deferred.

Table 1: Path B adapter modules. "Granularity" denotes whether the adapter's output is per-star (one row per star, pre-combined velocity) or per-epoch (one row per observation, fed through the multi-epoch combiner).

Adapter module	Galaxy / galaxies (LVDB key)	Granularity
`bruce2023`	`aquarius_2`	per-star
`chiti2022`	`grus_1`	per-epoch
`chiti2023`	`tucana_2`	per-epoch
`hansen2024`	`tucana_5`	per-epoch
`heiger2024`	`centaurus_1`	per-star
`ji2021`	`antlia_2`, `crater_2`	per-star
`kirby2015`	`pisces_2`	per-star
`koposov2015`	`horologium_1`	per-star
`koposov2018`	`hydrus_1`	per-star
`li2017`	`eridanus_2`	per-epoch
`li2018`	`carina_2`, `carina_3`	per-epoch
`simon2020`	`tucana_4`	per-epoch
`tan2025`	`leo_6`	per-star
`walker2009`	`carina_1`	per-star
`walker2015`	`reticulum_2`	per-star

2.3 Multi-epoch combination

Per-epoch catalogs are reduced to per-star arrays in `src/dwarfjeans/ingest/multi_epoch.py`; the dispatched combiner in `src/dwarfjeans/ingest/combiners/default.py` groups rows by `star_id` and applies inverse-variance weighting (IVW) under a per-paper `CombinePolicy` (`combiners/__init__.py`, frozen dataclass). Dispatch is keyed by `source_paper_bibcode` stamped into each adapter's `meta_extra`; per-paper handlers currently cover Li+2017 (Eri II), Li+2018 (Car II/III), Chiti+2022 (Gru I), Chiti+2023 (Tuc II), Simon+2020 (Tuc IV), and Hansen+2024 (Tuc V).

Default IVW (the "$\sigma_{\rm sys}$-as-statistical" path). Under the default convention, the published per-epoch error $\sigma_{\varepsilon,i}$ is fed directly into the IVW. For a star with N epochs,

$$\bar{v} = \frac{\sum_{i=1}^{N} v_i/\sigma_{\varepsilon,i}^2}{\sum_{i=1}^{N} 1/\sigma_{\varepsilon,i}^2}, \qquad \sigma_{\bar{v}} = \sqrt{\frac{1}{\sum_{i=1}^{N} 1/\sigma_{\varepsilon,i}^2} + \sigma_{\rm sys}^2}, \tag{4}$$

with $\sigma_{\rm sys} = 0$ for every per-paper handler under the current convention.

Bias from the default convention. Every UFD spectroscopy paper in the current ingest reports per-epoch errors that already include a systematic floor in quadrature, $\sigma_\varepsilon^2 = \sigma_{\rm stat}^2 + \sigma_{\rm sys}^2$. Feeding σ_ε into eq. (4) treats $\sigma_{\rm sys}$ as if it were statistical, so the IVW averages it down by $1/\sqrt{N}$ along with $\sigma_{\rm stat}$. The combined uncertainty is therefore biased low by approximately $\sigma_{\rm stat}/\sqrt{\sigma_{\rm stat}^2 + \sigma_{\rm sys}^2}$ — a $\sim$ 10–30% underestimate at typical $N = 2$–5 when $\sigma_{\rm sys} \sim \sigma_{\rm stat}$. The variability χ^2 p-values are correspondingly biased high. The bias is below the chain-noise floor for current $\sigma_{\rm los} \sim$ few-km s^{-1} inferences but is non-negligible for galaxies with ≤ 3 surviving members (e.g. Tuc V); see the `multi_epoch.py` module docstring and `docs/plan/per_paper_combiners.md`.

Strict deconvolution (opt-in). Setting `CombinePolicy.sigma_sys_kms` to a positive value routes the handler through `combine_star_strict` / `ivw_combine_strict` in `multi_epoch.py`. Per-epoch statistical components are recovered as $\sigma_{\mathrm{stat},i} = \sqrt{\sigma_{\varepsilon,i}^2 - \sigma_{\mathrm{sys}}^2}$, the IVW operates on $\sigma_{\mathrm{stat},i}$, and σ_{sys} is re-added in quadrature post-combine:

$$\bar{v}_{\mathrm{strict}} = \frac{\sum_{i=1}^{N} v_i/\sigma_{\mathrm{stat},i}^2}{\sum_{i=1}^{N} 1/\sigma_{\mathrm{stat},i}^2}, \qquad \sigma_{\bar{v},\,\mathrm{strict}} = \sqrt{\frac{1}{\sum_{i=1}^{N} 1/\sigma_{\mathrm{stat},i}^2} + \sigma_{\mathrm{sys}}^2}. \tag{5}$$

This is the textbook treatment of a fully correlated instrument systematic; no per-paper handler currently activates it.

2.4 Per-instrument zero-point offsets

`CombinePolicy.zero_point_offsets_kms` maps an instrument tag (read from the per-epoch `Inst` column) to an additive shift applied to V before the IVW:

$$v_i' = v_i + \delta_{\mathrm{zp}}(\texttt{Inst}_i). \tag{6}$$

A non-empty offset table requires an `Inst` column; a missing column or unknown tag raises (silent zero-offset fallback is explicitly forbidden). The reference instrument is defined by assigning offset zero. Chiti et al. (2023; Tucana II) is the only handler currently using non-zero offsets. With MIKE as reference, the `chiti2023` policy applies $\delta_{\mathrm{zp}} = \{\mathrm{MIKE}{=}\ 0,\ \mathrm{M2FS}{=}\ +2.5,\ \mathrm{IMACS}{=}\ +2.2,\ \mathrm{MagE}{=}\ +1.0\}\,\mathrm{km\,s^{-1}}$ per the inter-instrument offsets reported in that paper's §3.1 (byte-verified against the staged per-epoch table; see `docs/plan/per_paper_combiners.md`). Application shifts individual $\bar{v}$ values by up to $+2.5\,\mathrm{km\,s^{-1}}$ and reduces the proxy σ_{los} across the 19 Tuc II member stars from 4.02 to $3.88\,\mathrm{km\,s^{-1}}$ ($\approx 3\%$).

2.5 Variability flagging

Stars with $N \geq 2$ epochs are tested for velocity variability:

$$\chi^2 = \sum_{i=1}^{N} \frac{(v_i - \bar{v})^2}{\sigma_i^2}, \qquad \mathrm{dof} = N - 1, \tag{7}$$

with $\sigma_i = \sigma_{\varepsilon,i}$ under the default path and $\sigma_i = \sigma_{\mathrm{stat},i}$ under the strict path. The upper-tail p-value is compared against `CombinePolicy.p_threshold` (framework default 0.01): if $p < p_{\mathrm{threshold}}$, the star receives `var_flag=True`. Single-epoch stars are returned with $\chi^2 = p = \mathrm{NaN}$ and `var_flag=False`; no test is performed. The framework default of 0.01 matches Simon+2020 §3.4 explicitly and is consistent with the Li+2018 / Chiti+2023 hand-flag rules of thumb on inter-epoch Δv.

2.6 Per-paper choices summary

The canonical table of per-paper combiner settings is maintained in `docs/plan/per_paper_combiners.md`. All seven per-paper handlers currently set `CombinePolicy.sigma_sys_kms=0.` — not because the source papers omit a systematic floor, but precisely because every paper in the per-epoch set (Li+2017, Li+2018, Chiti+2022, Chiti+2023, Simon+2020, Hansen+2024) reports a per-instrument σ_{sys} in the range 0.5–1.2 $\mathrm{km\,s^{-1}}$ that is *already* added in quadrature into the published e_{RVel} column. Setting `sigma_sys_kms>0` would double-count. All handlers retain `p_threshold=0.`; only `chiti2023` populates a non-empty `zero_point_offsets_kms` mapping.

2.7 Star selection

Per-star catalogs from Stages 0b are reduced to the kinematic sample fed to Stage 1 by `select_jeans_stars` in `src/dwarfjeans/jeans/selection.py`, called automatically by `prepare_jeans_input`. Three cuts are applied in order:

1. **Membership**: $p_i > p_{\min}$ (default $p_{\min} = 0.5$). For Path A, p_i is the binary `Pmem-novar` from Paper I (1 = member with velocity variables already removed, 0 otherwise). For Path B, p_i is the per-paper adapter's published value. Catalogs whose p is uniformly 1 (already hard-cut upstream) skip the cut.

2. **Radial**: $R_i < r_{\rm cut} \equiv 2\, r_{1/2,\rm maj} \sqrt{1-\varepsilon} \cdot 4/3$, the sphericalised 3D Plummer half-mass radius scaled by 2. This convention is empirically tuned to reproduce Paper II Table A1 N across the Path A sample (17/22 galaxies within ± 2 stars; the five outliers — Leo I, Leo II, Sextans, Ursa Minor, Hercules — differ by ≤ 26 stars and trace to `Pmem-novar` snapshot drift, not to the cut definition). For galaxies with no published ellipticity (`ellipticity_missing=True`), $\varepsilon = 0$ is used and the missing-flag is recorded in the audit. The constant 4/3 is the conventional round approximation to the Plummer 3D/2D ratio and is kept distinct from the exact 1.30477 used by the derived $M_{1/2,\rm 3D}$ chain (Section 5); the $\sim$2.2% discrepancy reflects the empirical, not analytical, basis of the cut.

3. **Variability**: drop velocity-variable stars (`var_flag=True` or `Var=1.`). For Path A the variability removal is already folded into `Pmem-novar` upstream, so this cut is a no-op (defensive belt-and-braces).

The selection report (audited per-galaxy in `audit.json`) records `n_input`, `n_after_p`, `n_after_R`, `n_after_var`, `n_final`, `rhalf_major_pc`, `ellipticity`, `ellipticity_missing`, `rhalf_sph_3d_pc`, and `R_max_kpc`.

3 Stage 1: Spherical Jeans inference

The canonical extended treatment of the Jeans inference (motivation, parameter parameterizations, log-space numerical scheme, sampler configuration, MC validation plan) lives in `docs/plan/stage2.md` in the working plan; the high-level role in the broader pipeline is in `docs/plan/pipeline_overview.md`. The full algebraic derivation of the conditional Jeffreys prior $p(\ln\rho_s, \ln r_s \mid \beta, r_p, \text{data})$ used here lives in `docs/plan/jeffreys_jeans_derivation.md`; this section quotes the resulting formula but not the full derivation. Nuisance-prior conventions (split-normal targets, the symmetric-Gaussian interim) are in `docs/plan/uncertainty_conventions.md`. The Walker constant-σ baseline reported alongside (Section 3.5) is the plan's Stage 1 (`docs/plan/stage1.md`) — it's bundled here in the writeup because it shares the same per-galaxy driver.

3.1 Model

The pipeline models each dwarf as a spherically symmetric system in dynamical equilibrium, with an NFW dark-matter halo, a Plummer stellar tracer, and constant velocity anisotropy.

NFW halo. The dark-matter density profile is

$$\rho_{\rm DM}(r) = \frac{\rho_s}{(r/r_s)(1+r/r_s)^2}, \tag{8}$$

with ρ_s in $M_\odot\,\mathrm{kpc}^{-3}$ and r_s in kpc. The enclosed mass follows analytically:

$$M(r) = 4\pi\,\rho_s\,r_s^3\,g(r/r_s), \qquad g(x) = \ln(1+x) - \frac{x}{1+x}, \tag{9}$$

implemented in `src/dwarfjeans/jeans/solver.py` (`nfw_M`, `nfw_g`). For numerical stability $g(x)$ is evaluated via a Taylor series for $x < 10^{-3}$. No tidal-radius truncation is applied at this stage; the integration upper limit is set to $100\max(r_s, r_p)$, well into the Plummer r^{-5} tail.

Plummer tracer. The 3D stellar number density and projected surface density are

$$\nu(r) = \frac{3}{4\pi r_p^3}\left(1 + \frac{r^2}{r_p^2}\right)^{-5/2}, \qquad \Sigma(R) = \frac{1}{\pi r_p^2}\left(1 + \frac{R^2}{r_p^2}\right)^{-2}, \tag{10}$$

where r_p is the 3D Plummer scale (`plummer_nu`, `plummer_Sigma` in `solver.py`). For the Plummer profile the projected half-light radius equals r_p. In 7-parameter nuisance mode, the physical Plummer scale is derived per draw as

$$r_p = d\cdot\vartheta_{1/2}\cdot\frac{\pi}{180\times 60}\sqrt{1-\varepsilon}. \tag{11}$$

Anisotropy. Velocity anisotropy is parametrised by the constant $\beta \equiv 1 - \sigma_\theta^2/\sigma_r^2$, sampled through the symmetrised variable

$$\tilde\beta \in [-0.95,\,1), \qquad \beta = \frac{2\tilde\beta}{1+\tilde\beta}, \tag{12}$$

which maps the physical range $\beta \in (-\infty, 1)$ onto a bounded interval that is numerically convenient for nested sampling (`beta_tilde_to_beta` in `src/dwarfjeans/jeans/inference.py`). The tangential edge $\tilde\beta = -0.95$ corresponds to $\beta \approx -38$.

Jeans equation. The spherical Jeans equation for constant β,

$$\frac{1}{\nu}\frac{d(\nu\sigma_r^2)}{dr} + \frac{2\beta\,\sigma_r^2}{r} = -\frac{GM(r)}{r^2},$$

admits the integrating factor $r^{2\beta}$ and yields the closed-form integral solution (Pace & Strigari 2018, eqs. 3–7; hereafter P&S 2018):

$$\nu(r)\,\sigma_r^2(r) = r^{-2\beta}\int_r^\infty s^{2\beta}\,\nu(s)\,\frac{G\,M(s)}{s^2}\,ds. \tag{13}$$

This is integral J1 of the module docstring. Production evaluation uses a log-spaced grid in s with reverse cumulative log-sum-exp (`nu_sigma_r2_grid` in `solver.py`) to avoid float64 overflow at the extreme tangential edge where $2\beta \approx -76$.

Projection integral. The line-of-sight velocity dispersion at projected radius R is obtained from eq. (13) via the constant-β projection kernel (P&S 2018 eq. 4):

$$\Sigma(R)\,\sigma_{\rm los}^2(R) = 2\int_R^\infty \left(1 - \beta\,\frac{R^2}{r^2}\right)\frac{\nu(r)\,\sigma_r^2(r)\,r}{\sqrt{r^2 - R^2}}\,dr. \tag{14}$$

The square-root singularity at $r = R$ is regularised by the substitution $r^2 = R^2 + u^2$, giving a smooth integrand on $u \in [0, \infty)$ that is integrated by a per-R log-spaced trapezoidal rule with an explicit $u = 0$ endpoint (`Sigma_sigma_los2_grid` in `solver.py`). The public interface returns $\sigma_{\rm los}(R) = \sqrt{\Sigma\sigma_{\rm los}^2/\Sigma}$ (`sigma_los`).

3.2 Likelihood

Per-star Gaussian. Each star i has a measured line-of-sight velocity V_i, a combined measurement uncertainty $\sigma_{\varepsilon,i}$ (which already incorporates the instrument systematic floor from Stage 0b), a projected radius R_i, and a membership probability p_i. The total per-star variance is

$$s_i^2 \equiv \sigma^2_{\rm los}(R_i;\, \rho_s, r_s, \beta, r_p) + \sigma^2_{\varepsilon,i}, \tag{15}$$

and the per-star log-Gaussian is

$$\ln \mathcal{L}_i = -\tfrac{1}{2}\ln\!\left(2\pi s_i^2\right) - \frac{(V_i - \bar{V})^2}{2\, s_i^2}. \tag{16}$$

No additional intrinsic scatter term is added here; the systematic floor is already in $\sigma_{\varepsilon,i}$.

Membership-weighted pseudo-likelihood. Membership enters as an exponent on each per-star density (Walker et al. 2006 convention). The total log-likelihood summed over N stars is

$$\ln \mathcal{L} = \sum_{i=1}^{N} p_i\, \ln \mathcal{L}_i = \sum_{i=1}^{N} p_i \left[-\tfrac{1}{2}\ln(2\pi s_i^2) - \frac{(V_i - \bar{V})^2}{2\, s_i^2}\right], \tag{17}$$

implemented as `ll=float(np.sum(p*ln_li))` in `inference.make_loglike` and `inference.make_loglike_with_nuisances`. Stars with $p_i = 1$ (post-membership-cut samples) reduce eq. (17) to an ordinary unweighted sum; the general p_i-weighted form is used for catalogs that retain probabilistic membership. Stars below the membership threshold $p_{\rm min}$ are excluded by `prepare_jeans_input` before inference begins. The Fisher information of this pseudo-likelihood, used by the Jeffreys prior (Section 3.3), is derived in `docs/plan/jeffreys_jeans_derivation.md`.

Free parameters. The default 7-D inference vector (`marginalize_nuisances=True`, the production setting) is

$$\theta = \left(\bar{V},\ \log_{10} r_s,\ \log_{10}\rho_s,\ \tilde{\beta},\ d,\ \varepsilon,\ \vartheta_{1/2}\right),$$

where $\vartheta_{1/2}$ is the angular half-light radius (arcmin). The nuisances $(d, \varepsilon, \vartheta_{1/2})$ enter the likelihood not only through r_p (eq. 11) but also through the physical projected radius of each star,

$$R_i = d \cdot \Theta_i \cdot \frac{\pi}{180 \times 60}\ [\mathrm{kpc}], \tag{18}$$

where Θ_i is the angular separation from the galaxy centre. A larger distance pushes stars to larger physical radii and samples a different part of the dispersion profile. The hard constraint $r_s > r_p$ is enforced per draw inside the likelihood; violators return $\ln \mathcal{L} = -10^{300}$ and are rejected by the sampler. The fixed-geometry mode (`marginalize_nuisances=False`) holds the three nuisances at registry values and reduces θ to the 4-D vector $(\bar{V}, \log_{10} r_s, \log_{10}\rho_s, \tilde{\beta})$.

3.3 Priors

All prior bounds and family choices are in `src/dwarfjeans/jeans/priors.py`.

Halo base prior. Four base families for $(\ln \rho_s, \ln r_s)$ are registered under the key `prior_name`:

`uniform` linear-uniform on the physical values (r_s, ρ_s) with $r_s \in [10^{-2}, 10]$ kpc and $\rho_s \in [10^4, 10^{14}]\, M_\odot\,\mathrm{kpc}^{-3}$. The chain still records $(\log_{10} r_s, \log_{10} \rho_s)$.

`loguniform` uniform on $(\log_{10} r_s, \log_{10} \rho_s)$ over the same bounds: $\pi(\ln \rho_s, \ln r_s) = \mathrm{const}$.

`jeffreys` the loguniform base transform plus the Jeffreys log-determinant correction $\frac{1}{2} \ln D$ added to the log-posterior on every likelihood evaluation.

`satgen` a ΛCDM-conditioned prior built from the SatGen semi-analytic subhalo catalog with a Gaussian conditional on $\log_{10} \rho_s \mid \log_{10} r_s$ (Section 3.3).

`satgen_box` same SatGen catalog but uniform-in-log inside the per-bin envelope of populated halos rather than a Gaussian conditional (Section 3.3).

`satgen_shmr` per-dwarf SatGen prior obtained by reweighting the same halo catalog with precomputed stellar-to-halo-mass-relation (SHMR) weights from `SatGen_Dwarf`; same Gaussian-conditional functional form as `satgen` but with weighted bin moments and bin edges (Section 3.3).

The production default is `jeffreys` (`scripts/run_production.py`).

Jeffreys log-determinant. The dispersion $\sigma^2_{\mathrm{los}}(R_i)$ is linear in ρ_s at fixed (r_s, β); its $\ln r_s$-derivative satisfies $\partial\sigma^2_{\mathrm{los}}/\partial \ln r_s = \sigma^2_{\mathrm{los}}\, T_i$ with auxiliary shape factor $T_i = 3 - \mathcal{Q}(R_i)/\mathcal{P}(R_i)$ (`docs/plan/jeffreys_jeans_derivation.md`, Steps 2–3). With per-star weights $\tilde{w}_i = A_i^2/(A_i+\sigma^2_{\varepsilon,i})^2$ and $A_i = \sigma^2_{\mathrm{los}}(R_i)$, and membership-weighted sum $S_0 = \sum_i p_i \tilde{w}_i$, the 2×2 Fisher matrix in $(\ln \rho_s, \ln r_s)$ has determinant

$$D = S_0 \sum_i p_i\, \tilde{w}_i\, (T_i - \bar{T})^2, \qquad \bar{T} = \frac{\sum_i p_i\, \tilde{w}_i\, T_i}{S_0}, \tag{19}$$

in the variance form (algebraically equivalent to $S_0 S_2 - S_1^2$ but free of catastrophic cancellation). The Jeffreys correction is $\frac{1}{2} \ln D$. When $D \le 0$ or non-finite (all stars at the same R_i/r_s, so r_s is unidentifiable), the correction returns $-\infty$ and the draw is rejected. The correction requires T_i, obtained by calling `jeans.sigma_los_with_T` instead of `jeans.sigma_los`; this approximately doubles the cost of each likelihood evaluation.

SatGen-conditioned prior. The `satgen` family draws $(\log_{10} r_s, \log_{10} \rho_s)$ from the ΛCDM-with-tidal-stripping subhalo catalog `m12res8_10k_Diemer+scatter_sim.h5` in the companion `SatGen_Dwarf` repository ($\sim 2.4\times 10^6$ surviving and disrupted subhalos). Each halo's $(v_{\max}, r_{\max})$ is mapped to NFW (r_s, ρ_s) analytically: the peak of $v_c(r)$ for an NFW profile occurs at $x_{\max} \equiv r_{\max}/r_s \approx 2.16258$, the root of $\ln(1+x) = x(1+2x)/[2(1+x)^2]$, so

$$r_s = r_{\max}/x_{\max} = 0.46241\, r_{\max}, \tag{20}$$

$$\rho_s = \frac{x^3_{\max}}{4\pi G\, \mu(x_{\max})} \left(\frac{v_{\max}}{r_{\max}}\right)^2 = 1.72126\, \frac{(v_{\max}/r_{\max})^2}{G}, \tag{21}$$

with $\mu(x) = \ln(1+x) - x/(1+x)$, $v_{\max}$ converted from $\mathrm{kpc\,Gyr^{-1}}$ (SatGen's native units) to $\mathrm{km\,s^{-1}}$, and G in $(\mathrm{km\,s^{-1}})^2\,\mathrm{kpc}/M_\odot$. Halos outside the registry box $\log_{10}(r_s/\mathrm{kpc}) \in [-2, 1]$ are discarded ($\sim 1.4\times 10^3$ rows of $\sim 2.4\times 10^6$).

We then build a one-dimensional inverse-CDF table for the SatGen marginal $p(\log_{10} r_s)$ on a 1024-node grid, and tabulate $\mu(\log_{10} r_s)$ and $\sigma(\log_{10} r_s)$ as the per-bin sample mean and sample standard deviation (`ddof=1`) of $\log_{10}\rho_s$ in 300 *quantile-cut* bins (`scripts/build_satgen_prior_table.py`; cached at `data/satgen_prior/m12res8_diemer_scatter.npz`). Bin edges are placed at equally-spaced quantiles of the empirical $\log_{10} r_s$ distribution, so every bin contains $\sim N_{\text{total}}/N_{\text{bins}} \approx 8\times10^3$ halos by construction; no threshold-based extrapolation or moment-flooring is needed. The prior transform draws

$$\log_{10} r_s \sim F^{-1}_{\text{SatGen}}(u_1), \tag{22}$$

$$\log_{10}\rho_s \mid \log_{10} r_s \sim \mathcal{N}\big(\mu(\log_{10} r_s),\, \sigma(\log_{10} r_s)\big), \tag{23}$$

with $\log_{10}\rho_s$ unbounded apart from a $u_2 \in [10^{-15},\, 1-10^{-15}]$ numerical guard (a $\pm7.94\sigma$ envelope on the conditional Gaussian whose excluded mass is $\sim 10^{-15}$). The conditional tails extend beyond the registry box on ρ_s, but the per-bin scatter is $\sigma \lesssim 0.5$ so the box edge is reached only on $\lesssim 10^{-9}$ of draws in the bulk. An optional `log10_rs_min` kwarg re-truncates F_{SatGen} to [`log10_rs_min`, 1] and renormalises; it is honoured by the 4-D builder `make_satgen_prior_transform` but *not* by the 7-D `make_satgen_prior_transform_with_nuisances` used in production (hard-coded `None`). V and $\tilde{\beta}$ priors are unchanged.

The lognormal-in-ρ_s approximation is justified by numerical diagnostics on the catalog itself, regeneratable via `scripts/build_satgen_prior_table.py`: pooled standardised residuals across the 2.4M halos have skewness +0.13 and excess kurtosis +0.21, with the corresponding QQ relation tracking $\mathcal{N}(0,1)$ to within $\lesssim 0.1$ in z across $\pm4\sigma$. With quantile-cut bins every bin has $\sim 8\times10^4$ halos, so per-bin (μ,σ) estimates carry negligible sampling noise and no bin sits at the unfaithful edge. Per-bin μ, σ, skewness, excess kurtosis, and counts are stored alongside the lookup arrays in `data/satgen_prior/m12res8_diemer_scatter.npz` (`skew_per_bin`, `kurt_per_bin`, `n_per_bin`); the top-level `metadata` JSON records the pooled moments and the source-file SHA. A 2D KDE form is the natural escalation if a future analysis becomes sensitive to higher-moment structure.

SatGen envelope (box) prior. The `satgen_box` family shares the same SatGen catalog and the same $(v_{\max}, r_{\max}) \to (r_s, \rho_s)$ mapping (eq. 20, 21) and the same per-bin machinery, but replaces the Gaussian conditional in eq. 23 with a uniform-in-log envelope. For each of the same 300 quantile-cut $\log_{10} r_s$ bins we record the empirical extrema

$$\rho_{\text{lo}}(k) = \min_{i\in\text{bin}\,k} \log_{10}\rho_{s,i}, \quad \rho_{\text{hi}}(k) = \max_{i\in\text{bin}\,k} \log_{10}\rho_{s,i}. \tag{24}$$

Quantile-cut binning guarantees every bin holds $\sim 8\times10^4$ halos, so the empirical $(\rho_{\text{lo}}, \rho_{\text{hi}})$ are well-defined in every bin and the prior marginal on $\log_{10} r_s$ is uniform per unit support length across the bins,

$$\log_{10} r_s \sim \text{Uniform}\Big(\bigcup_{k:\text{good}} [\,\text{edge}_k,\, \text{edge}_{k+1}\,]\Big), \tag{25}$$

$$\log_{10}\rho_s \mid \log_{10} r_s \sim \text{Uniform}\big(\rho_{\text{lo}}(k),\, \rho_{\text{hi}}(k)\big), \tag{26}$$

where k is the bin containing the drawn $\log_{10} r_s$. Hard truncation: no draws land outside the empirical SatGen support, so no Gaussian-tail leakage past the registry box and no extrapolation into bins with zero halos. V and $\tilde{\beta}$ priors are unchanged. The new lookup fields `bin_edges_log10_rs`, `rho_lo_log10_rhos`, `rho_hi_log10_rhos`, and the scalars `log10_rs_min_data`, `log10_rs_max_data` are stored in the same cache file alongside the Gaussian-prior fields. The `satgen_box` family is intended as the closer point of comparison to importance-reweighted SatGen analyses, which carry the empirical halo distribution rather than a per-bin Gaussian fit.

SHMR-weighted SatGen prior. The satgen_shmr family is the per-dwarf refinement of satgen: the same Diemer halo catalog (eq. 20, 21) is reweighted by precomputed importance weights w_i tied to a specific dwarf, producing one cached lookup table per (SHMR, dwarf) pair instead of the single galaxy-agnostic table used by satgen. The weight files live in the companion SatGen_Dwarf repository under data/additional/weights_gc/Diemer/<dwarf>.npz and are row-aligned to the halo catalog. The upstream file contains ten precomputed SHMRs; the production CLIs (run_production.py, run_batch.py, plot_posteriors.py) currently wire four of them through the --shmr flag: Fattahi et al. (2018), Moster et al. (2018), Danieli et al. (2023, constant-slope variant), and Kim et al. (2024). The remaining six (Behroozi+13, Moster+13, Rodriguez-Puebla+17, Behroozi+19, Munshi+21, and the growing-slope Danieli+23 variant) are available in the build script but not exposed at the CLI by default.

Each weight w_i is the product of an SHMR likelihood term — a Bayesian score of the halo's MC-sampled stellar mass against the dwarf's observed $M_\star$ distribution — and a geometric gate that zeroes any halo whose galactocentric distance lies outside a factor of two (in either direction) of the dwarf's observed value. We consume the precomputed weight arrays directly; the SHMR formulae, scatter conventions, and the geometric cut are defined in SatGen_Dwarf/python/halo_weights.py and not re-derived here. The geometric gate is dominant: across the 39-dwarf study sample it zeroes between 28% and 94% of the catalog, leaving $\sim 10^5$–10^6 non-zero halos per dwarf. Zero-weight halos are discarded before any binning, so the weighted moments and quantile-cut bin edges operate strictly on the geometrically- and SHMR-consistent sub-population.

The functional form of the prior is identical to satgen (eq. 22–23), but every ingredient is computed with weights. Bin edges in $\log_{10} r_s$ are placed at uniform quantiles of cumulative weight,

$$F^{(w)}(x) = \frac{\sum_{i:\, x_i \le x} w_i}{\sum_i w_i}, \qquad \mathrm{edge}_k = (F^{(w)})^{-1}\Big(\tfrac{k}{N_{\rm bins}}\Big), \tag{27}$$

and per-bin moments use the reliability-weighted mean and the reliability-weight Bessel-corrected variance,

$$\mu_k = \frac{\sum_{i\in k} w_i\, y_i}{\sum_{i\in k} w_i}, \quad \sigma_k^2 = \frac{\sum_{i\in k} w_i\,(y_i-\mu_k)^2}{\sum_{i\in k} w_i - \sum_{i\in k} w_i^2 / \sum_{i\in k} w_i}, \tag{28}$$

where $y_i \equiv \log_{10}\rho_{s,i}$. Eq. 28 reduces to np.std(...,ddof=1) when all weights are equal, and inflates the denominator by the effective sample size for unequal weights, matching the satgen convention.

We compute the global effective sample size $\mathrm{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$ for each dwarf and default to $N_{\rm bins} = 300$; if $\mathrm{ESS}/N_{\rm bins} < 30$ we adaptively reduce $N_{\rm bins}$ to $\max(50, \lfloor \mathrm{ESS}/30 \rfloor)$ so each bin retains at least ~ 30 effective halos. In the Fattahi+18 build of the 39-dwarf sample only Hydrus I triggered the fallback ($N_{\rm bins} = 292$); ESS ranged from $\sim 9\times10^3$ to $\sim 1.3\times10^5$. Tables and per-dwarf metadata (n_halos_nonzero, zero_weight_fraction, ess, n_bins) are cached at data/satgen_prior/<shmr>/<lvdb_key>.npz; the builder is scripts/build_satgen_shmr_prior_tables.py and the loader shares the satgen code path (make_satgen_shmr_prior_transform in src/dwarfjeans/jeans/priors.py is a thin wrapper that resolves the per-dwarf table path and delegates to make_satgen_prior_transform). The satgen_shmr prior has no Walker constant-σ analog on $\sigma_{\rm los}$, so the companion in run_production.py falls back to jeffreys for that inference, as for satgen/satgen_box.

Anisotropy and systemic priors. $\tilde\beta$ is flat on $[-0.95, 1)$ (eq. 12); the open upper bound avoids the radial-orbit singularity at $\beta = 1$. $\bar V$ is flat on $[V_{\rm centre} - 10,\, V_{\rm centre} + 10]\,\mathrm{km\,s^{-1}}$, where $V_{\rm centre}$ is the inverse-variance-weighted mean of the post-selection velocities, $V_{\rm centre} = \sum_i V_i/\sigma_{\varepsilon,i}^2 / \sum_i 1/\sigma_{\varepsilon,i}^2$.

Centering on the data rather than on the registry `vlos_systemic_kms` avoids stale/sample-mismatch failures (Pegasus III hit the lower wall of a registry-centered prior in QA).

Nuisance priors (7-D mode). When `marginalize_nuisances=True`, three additional parameters are sampled with independent Gaussian priors derived directly from the registry. The interim symmetric-Gaussian form below matches the current implementation; the design target (split-normal priors for asymmetric LVDB error fields) and the rationale for both choices are in `docs/plan/uncertainty_conventions.md`.

$$d \sim \mathcal{N}(\mu_d,\, \sigma_d), \tag{29}$$

$$\varepsilon \sim \mathcal{N}(\mu_\varepsilon,\, \sigma_\varepsilon) \text{ truncated to } [0, 1), \tag{30}$$

$$\vartheta_{1/2} \sim \mathcal{N}(\mu_\vartheta,\, \sigma_\vartheta), \tag{31}$$

with $\sigma_d = (\ln 10/5)\, d\, \sigma_\mu$ from the registry distance modulus and its uncertainty (`run_production._registry_nuisance_priors`). For ellipticity-missing galaxies, σ_ε is set to half the registry upper limit.

3.4 Sampler

Algorithm. Static nested sampling with DYNESTY (Speagle 2020); bounding method `multi`, proposal method `rwalk`. Production defaults: $N_{\rm live} = 500$ and stopping criterion $\Delta \ln \mathcal{Z} < 0.1$. The random state is seeded via `np.random.default_rng(rseed)` with default `rseed=0`.

Post-processing. After `sampler.run_nested`, normalised log-weights are $\log w_k = \log \mathrm{wt}_k - \ln \mathcal{Z}$, and `dynesty.utils.resample_equal` produces an equal-weight chain `samples_eq` from which all downstream quantities are derived. The driver returns the raw and equal-weight samples, log-weights, $\ln \mathcal{Z}$ and its uncertainty, and the parameter-name tuple.

Derived chains. The production driver `scripts/run_production.py` pushes `samples_eq` through: per-sample r_p and the radii $R_{1/2,2\rm D} = r_p$ and $r_{1/2,3\rm D} = 1.30477\, r_p$; NFW enclosed masses $M(R_{1/2,2\rm D})$ and $M(r_{1/2,3\rm D})$ via `jeans.nfw_M`; $\sigma_{\rm los}$ at $R_{1/2,2\rm D}$ (thinned to 2000 draws); and J/D chains at four angles each (thinned to 500 draws, Section 5.4). Outputs land in `results/production/<lvdb_key>/<prior>/` (overwritten by each run) as `posterior_samples.npz`, a per-quantity 16/50/84-percentile `summary.csv`, and an `audit.json` recording registry inputs, selection policy, and dynesty configuration.

3.5 Walker constant-σ baseline

Alongside the Jeans posterior, the production driver also evaluates a radius-independent Walker (2006) dispersion as a model-free cross-check. This is the canonical comparison number quoted across the dwarf-galaxy literature (P&S 2018, Walker 2006), so reporting it in the same `summary.csv` as the Jeans result makes paper-vs-paper checks one-line.

Likelihood. A single global σ replaces the radial profile $\sigma_{\rm los}(R)$. The per-star membership-weighted Gaussian matches Eq. (17) stripped of the R-dependence ($s_i^2 \to \sigma^2 + \varepsilon_i^2$ instead of Eq. (15)):

$$\ln L_i \;=\; p_i \Big[-\tfrac{1}{2} \ln\big[2\pi(\sigma^2 + \varepsilon_i^2)\big] - \tfrac{1}{2}\, \frac{(V_i - V_{\rm sys})^2}{\sigma^2 + \varepsilon_i^2} \Big]. \tag{32}$$

Inference is on the two parameters $(V_{\rm sys}, \sigma)$ only; no halo parameters and no nuisances enter.

Priors. Uniform on $V_{\rm sys} \in [V_{\rm centre} \pm 10 \text{ km s}^{-1}]$ ($V_{\rm centre}$ is the IVW mean of the post-selection velocities, matching the Stage 2 prior; see the anisotropy-and-systemic-priors paragraph). On σ we use the *proper* (Fisher-determinant) Jeffreys prior, computed at each grid point from the diagonal Fisher information $I_{\bar V \bar V} = 1/\sigma_i^2$ and $I_{\sigma\sigma} = 2\sigma^2/\sigma_i^4$ with $\sigma_i^2 = \sigma^2 + \varepsilon_i^2$:

$$p_J(\sigma) \;\propto\; \sigma \sqrt{\Big[\sum_i p_i/\sigma_i^2\Big]\Big[\sum_i p_i/\sigma_i^4\Big]}, \tag{33}$$

which is $\bar V$-independent and truncated to $\log_{10}\sigma \in [-2, 2]$ (i.e. $\sigma \in [0.01, 100]$ km s^{-1}). The proper Jeffreys form is preferred over the bare log-uniform because the latter piles posterior mass at $\sigma \ll \varepsilon_i$ where the likelihood is σ-independent.

Implementation and reporting. The 2D posterior on $(V_{\rm sys}, \sigma)$ is evaluated on a 400×400 grid in `constant_sigma_inference`. Marginals on $V_{\rm sys}$ and σ are integrated by trapezoid; the 16/50/84 percentiles are read off the cumulative. The driver records the σ marginal as `sigma_los_walker_kms` in `summary.csv`. Runtime is sub-second per galaxy and is independent of the Jeans inference.

4 Stage 2: Population Monte Carlo calibration

The MC test design (truth grids, realisation count, bias / coverage metrics, rerun procedure) is documented in `docs/plan/README_mc_test.md`; the calibration role within the pipeline is in `docs/plan/stage2.md` ("MC recovery test" section).

4.1 Motivation

Before applying the Jeans inference to real kinematic data, the full chain — mock generation, Jeans projection, likelihood, sampler, posterior summarisation — is validated end-to-end on synthetic catalogs at known truth. Unit tests verify the mathematics of individual components; the population MC verifies that the complete chain is unbiased and well-calibrated on realistic inputs.

4.2 Mock galaxy generator

The mock generator is implemented in `src/dwarfjeans/mocks/galaxy.py`. Each realisation produces a per-star catalog $(R_i, V_i, \sigma_{\varepsilon,i}, p_i)$ matching the schema of the real kinematic data; the truth model is an NFW halo (r_s, ρ_s) with a Plummer tracer of scale r_p and constant anisotropy β.

Projected radii. Stars are drawn from the 2D Plummer surface density $\Sigma(R) \propto (1 + R^2/r_p^2)^{-2}$ via inverse-CDF sampling; the CDF $F(< R) = R^2/(R^2 + r_p^2)$ inverts analytically:

$$R_i = r_p \sqrt{\frac{u_i}{1 - u_i}}, \qquad u_i \sim \text{Uniform}(0, 1). \tag{34}$$

Draws with $R_i > 5\, r_p$ are rejected and redrawn, matching the membership-cut footprint of real spectroscopic surveys.

Line-of-sight velocities. For each R_i, $\sigma_{\rm los}(R_i)$ is evaluated at truth and

$$V_i \sim \mathcal{N}\big(V_{\rm sys},\, \sigma^2_{\rm los}(R_i) + \sigma^2_{\varepsilon,i}\big). \tag{35}$$

Membership probabilities are set to $p_i = 1$ (post-membership-cut).

4.3 Asimov realisation

A single Asimov realisation (`src/dwarfjeans/mocks/galaxy.make_asimov_galaxy`; driver `tests/integration/run_ufd_population.py--asimov`) isolates prior and projection bias from sampling noise. Star positions are placed at the equal-probability stratified midpoints of the truncated Plummer CDF on $R \in [0, 5\,r_p]$:

$$u_i = \frac{i - \frac{1}{2}}{N}, \quad R_i = r_p \sqrt{\frac{u_i\,F_{\max}}{1 - u_i\,F_{\max}}}, \qquad i = 1, \ldots, N, \tag{36}$$

where $F_{\max} = (5r_p)^2/[(5r_p)^2 + r_p^2]$. The Asimov likelihood replaces the squared velocity residual by its expectation $\sigma^2_{\mathrm{tot,truth}}(R_i)$ under the truth model, so the MLE sits exactly at truth. V_{sys} does not enter the Asimov likelihood and its posterior equals the prior; it is flagged `prior_only` and excluded from z-score diagnostics.

4.4 Population MC test

Implemented in `tests/integration/run_ufd_population.py`. Fifteen independent realisations are generated at the same truth:

$$\begin{array}{ll} r_s = 0.3\,\mathrm{kpc} & \rho_s = 3 \times 10^8\,M_\odot\,\mathrm{kpc}^{-3} \\ r_p = 0.05\,\mathrm{kpc} & \beta = 0 \\ N_\star = 30 & \sigma_\varepsilon = 2\,\mathrm{km\,s^{-1}} \end{array}$$

Each realisation uses a distinct random seed. Inference uses `nlive`=300, `dlogz`=0.5, log-uniform prior, and uniform sampling (`sample="unif"`) for speed — distinct from the production driver, which uses `sample="rwalk"` with the same `multi`-ellipsoidal bounds. Calibration diagnostics are computed for each parameter $\theta \in \{\bar{V},\, \log_{10} r_s,\, \log_{10} \rho_s,\, \tilde{\beta},\, \log_{10}(\rho_s r_s^3)\}$:

- asymmetric z-score $z = (\mathrm{median} - \theta_{\mathrm{truth}})/\sigma_\pm$,
- 68% and 95% credible-interval coverage,
- Kolmogorov–Smirnov p-value of $\{z\}$ against $\mathcal{N}(0, 1)$,
- mean z, std(z), and median bias.

Calibration results (Jeffreys prior, current default). The 15-realisation MC was rerun with `prior_name="jeffreys"` to match the production default (`run_ufd_population.py`). Per-parameter diagnostics:

Parameter	mean(z)	std(z)	med. bias	cov$_{68\%}$	cov$_{95\%}$	KS p
$\bar{V}$	-0.08	0.93	-0.08	87%	93%	0.75
$\log_{10} r_s$	$+0.72$	0.60	-0.65	60%	100%	0.0002
$\log_{10} \rho_s$	-0.77	0.63	$+1.10$	60%	100%	0.0003
$\tilde{\beta}$	$+0.17$	0.81	$+0.00$	87%	100%	0.73
$\log_{10}(\rho_s r_s^3)$	$+0.65$	0.56	-0.86	67%	100%	0.0002
$\log_{10} M(r_{1/2,\mathrm{2D}})$	-0.83	0.82	$+0.13$	53%	100%	0.013
$\log_{10} M(r_{1/2,\mathrm{3D}})$	-0.61	0.79	$+0.07$	60%	100%	0.071

The well-constrained derived quantity remains $M(r_{1/2,3\mathrm{D}})$ (typical 1σ width ≈ 0.12 dex, smaller than the loguniform value of 0.15 dex), but under Jeffreys it picks up a ~ 0.07 dex high-side median bias and a tighter-than-nominal z-distribution ($\mathrm{std}(z) = 0.79$, $\mathrm{cov}_{68\%} = 60\%$, KS $p = 0.07$). The individual halo parameters $\log_{10} r_s$, $\log_{10} \rho_s$, $\log_{10}(\rho_s r_s^3)$ are now meaningfully biased ($|\text{med. bias}| =$ 0.6–1.1 dex) with strongly non-normal z-distributions (KS $p < 10^{-3}$).

This is a marked departure from the historical loguniform-prior result, where $M(r_{1/2,3\mathrm{D}})$ was recovered with median bias < 0.01 dex, $\mathrm{std}(z) \approx 1.1$, and KS $p \approx 0.7$. The Jeffreys term shifts the joint posterior peak toward higher ρ_s / lower r_s in the small-x NFW regime relevant for UFDs ($r_{1/2}/r_s \approx 0.2$), where the data only constrain the enclosed-mass combination $M \propto \rho_s r_s r^2$. The inflated peak shifts $M(r_{1/2})$ slightly high and tightens the marginal posterior beyond the realisation-to-realisation spread, producing the observed under-dispersion.

Prior note. The 15-realisation MC is now run under the production prior (`prior_name="jeffreys"`) to keep the calibration aligned with production. The historical log-flat numbers (median bias < 0.01 dex, $\mathrm{std}(z) \approx 1.1$, KS $p \approx 0.7$) remain accessible by overriding to `prior_name="loguniform"`.

5 Stage 3: J- and D-factor extraction

The canonical extended treatment of the J/D-factor pipeline (the chain-push design, tidal-radius source choice, MC recovery plan, and deviations from P&S 2018) lives in `docs/plan/stage3.md`; this section summarizes the as-implemented integrals and reporting angles.

5.1 Integral definitions

The J- and D-factors quantify the line-of-sight integral of ρ^2 and ρ, respectively, over a solid angle $\Delta\Omega(\theta_{\max})$ centred on the target (`src/dwarfjeans/jd/factors.py`):

$$J(\theta_{\max}) = \int_{\Delta\Omega} \int_{\text{l.o.s.}} \rho^2\big(r(\ell,\Omega)\big)\, d\ell\, d\Omega, \tag{37}$$

$$D(\theta_{\max}) = \int_{\Delta\Omega} \int_{\text{l.o.s.}} \rho\big(r(\ell,\Omega)\big)\, d\ell\, d\Omega. \tag{38}$$

In the small-angle approximation (valid for $\theta_{\max} \lesssim 1^\circ$), $R = d\,\theta$ and $d\Omega \approx R\,dR/d^2$, giving

$$J(\theta_{\max}) = \frac{2\pi}{d^2} \int_0^{R_{\max}} R\, I_2(R;\, r_t)\, dR, \tag{39}$$

$$D(\theta_{\max}) = \frac{2\pi}{d^2} \int_0^{R_{\max}} R\, I_1(R;\, r_t)\, dR, \tag{40}$$

where $R_{\max} = \min(d\,\theta_{\max},\, r_t)$ and the line-of-sight column at impact parameter R for an NFW halo with hard tidal truncation at r_t is

$$I_n(R;\, r_t) = 2 \int_R^{r_t} \frac{\rho^n(r)\, r}{\sqrt{r^2 - R^2}}\, dr, \qquad n = 1, 2. \tag{41}$$

The integrable $1/\sqrt{r^2 - R^2}$ singularity is removed by $r^2 = R^2 + u^2$, giving a smooth integrand on $u \in [0, u_{\max}(R)]$ with $u_{\max}(R) = \sqrt{r_t^2 - R^2}$. Internally the pipeline works in $(M_\odot, \mathrm{kpc})$ and converts to P&S 2018 reporting units ($\mathrm{GeV}^2/\mathrm{cm}^5$ for J, $\mathrm{GeV}/\mathrm{cm}^2$ for D) at output.

5.2 Numerical implementation

The u-grid is log-spaced from $10^{-6}\,r_s$ to $u_{\max}(R)$ with an explicit $u = 0$ endpoint prepended (where the integrand is largest); trapezoidal quadrature is applied. The outer R-grid is log-spaced from $10^{-6}\,r_s$ to $R_{\max}$ with the on-axis point $R = 0$ excluded from the log grid (where $I_n(0)$ hits $\rho(0) = \infty$ for an NFW); the integrand $R\cdot I_n(R)$ is zero at $R = 0$, so an explicit $(R = 0, \text{integrand} = 0)$ endpoint is prepended. Production defaults from `J_D_factors` are $n_R = 128$, $n_u = 512$, calibrated against `scipy.integrate.quad` to better than 10^{-3} relative error over the MW-dwarf parameter range. `summarize_jd` (used by population-MC summarisation) defaults to $n_R = 64$, $n_u = 128$ and thins the chain to 500 samples; `analyze_asimov.py` runs at $n_R = 96$, $n_u = 256$.

5.3 Tidal angle and reporting angles

The characteristic integration angle for J follows the Wolf et al. (2010) critical-angle convention:

$$\alpha_c = \frac{2\,r_{1/2,3\mathrm{D}}}{d}, \tag{42}$$

computed via `alpha_c_radians` in `factors.py`. Each chain is evaluated at four angles:

$$J: \quad \theta_{\max} \in \{0.1^\circ,\, 0.2^\circ,\, 0.5^\circ,\, \alpha_c\}, \tag{43}$$

$$D: \quad \theta_{\max} \in \{0.1^\circ,\, 0.2^\circ,\, 0.5^\circ,\, \alpha_c/2\}. \tag{44}$$

The $\alpha_c/2$ choice for the D natural angle reflects the shallower projected ρ profile relative to ρ^2. In nuisance-marginalised mode α_c becomes a per-draw quantity; the fixed angles vary only through the sampled d.

5.4 Chain push

The Stage 1 equal-weight chain $(\bar{V}^{(k)},\, \log_{10} r_s^{(k)},\, \log_{10} \rho_s^{(k)},\, \tilde{\beta}^{(k)})$ is processed sample-by-sample in `scripts/run_production.py` via per-draw, per-angle calls to `J_D_factors` (`factors.py`). Per draw: (r_s, ρ_s) are extracted; d comes from the nuisance chain (production) or is held at truth (MC validation); the tidal radius r_t is computed per draw from the sampled halo and galactocentric distance (Section 5.5). `J_D_factors` returns $(J^{(k)}, D^{(k)})$ in internal units. The chains are summarised per galaxy as median + q_{16}/q_{84} + 1D-KDE MAP.

5.5 Tidal radius

The line-of-sight truncation r_t is computed per posterior draw in `src/dwarfjeans/jd/tidal.py` from the Tormen, Diaferio & Syer (1998) / Springel et al. (2008) radial tidal-radius condition:

$$r_t = D\left[\frac{M_{\mathrm{sub}}(< r_t)}{\big(\kappa - \mathrm{d}\ln M_{\mathrm{host}}/\mathrm{d}\ln r\big)\,M_{\mathrm{host}}(< D)}\right]^{1/3}, \tag{45}$$

with $\kappa = 2$ (radial tidal field; the Tormen/Springel convention, default) or $\kappa = 3$ (circular-orbit Jacobi/Hill, $\to D\,(m/3M)^{1/3}$ for a point-mass host), a swappable knob. $M_{\mathrm{sub}}(< r_t) = 4\pi\rho_s r_s^3\, g(r_t/r_s)$ is the NFW subhalo mass; the slope and the host enclosed mass are evaluated at the satellite's galactocentric distance D. Equation (45) is implicit in r_t and is solved with a one-dimensional bracketed root find (`tidal_radius`). It depends only on $(r_s, \rho_s, D, \text{host})$ — not on the kinematics — so it is computed for every dwarf, σ_{los}-resolved or not.

The host is the SatGen `m12res8-Diemer` Milky-Way halo ($M_{\rm vir} = 10^{12}\,M_\odot$, $R_{\rm vir} = 258.9\,$kpc, concentration $c \simeq 11.5$; NFW, $z = 0$), matching the satgen prior catalog. For an NFW, $\mathrm{d}\ln M/\mathrm{d}\ln r = h(x)/g(x)$ with $x = r/r_s$ (since $x\,g'(x) = x^2/(1+x)^2 = h(x)$), so the host log-slope reuses `nfw_g`/`nfw_h` from `solver.py`. The galactocentric distance is obtained from the registry (α, δ) and the heliocentric distance chain via astropy's default `Galactocentric` frame (`galactocentric_distance`). For a typical UFD this gives $r_t \sim 2$–$7\,$kpc, far outside the projected radius of the reporting angles — so J (set by ρ^2 near the centre) is essentially unchanged from the former $r_t = 1\,$kpc placeholder, while the more extended D (set by ρ) rises by a few percent.

5.6 Angular containment

Alongside the reporting-angle factors, the pipeline reports the posterior on the *95% containment angle* $\theta_{95}(J)$ — the half-angle of the cone containing 95% of the J-factor integrated out to r_t. This is computed by `j_containment` in `factors.py`, ported from the validated `compute_J_and_containment` of the companion SatGen analysis. Unlike `J_D_factors`, which uses the small-angle approximation $R = d\,\theta$, the containment integral uses the *exact* line-of-sight geometry,

$$r(\ell, \theta) = \sqrt{(\ell - d)^2 + 4\,\ell\,d\,\sin^2(\theta/2)}, \tag{46}$$

(`los_radius`; the half-angle form is identical to the law of cosines but free of catastrophic cancellation), valid to $\theta = \pi/2$, because θ_{95} reaches several tenths of a degree to $\sim 1^\circ$ for nearby dwarfs — beyond the small-angle regime. The NFW is truncated hard at r_t, so the angular profile self-limits at $\theta_{\rm max} = \arcsin(r_t/d)$ and the "total" J is $J(< r_t)$. The cumulative angular profile $\mathrm{d}J/\mathrm{d}\theta$ is built by deterministic Gauss–Legendre line-of-sight quadrature (closest-approach substitution clustering nodes where ρ peaks), and θ_{95} is read off the normalised cumulative; the $\theta \to 0$ cusp limit ($\mathrm{d}J/\mathrm{d}\theta \to$ const for an NFW $\gamma = 1$ inner slope) is used in place of the $0 \cdot \infty$ on-axis node. The same routine returns $J(< 0.5^\circ)$, which agrees with the small-angle `J_D_factors` value to < 0.01 dex; the full $J(< r_t)$ matches an independent `scipy.integrate.dblquad` reference to < 0.01 dex.

5.7 MC recovery

The 15 mocks of Section 4.4 are pushed through J/D with $d = 30\,$kpc and $r_t = 1\,$kpc held at truth (synthetic data carry no host-orbit information for a Springel et al. 2008 r_t).

Population-level J/D diagnostics are pending under the Jeffreys-prior MC. The historical numbers below were produced by the former `run_jd_summary.py` tool under the loguniform prior; the current `run_ufd_population.py` computes the halo recovery only. Reinstating per-realisation J/D summaries (via a `summarize_jd` call inside the MC loop, writing `ufd_pop_jd_diagnostics.json`) is a follow-up. Historical loguniform numbers, retained as a baseline: **D-factor recovery was clean** at all four angles (median bias ≤ 0.03 dex, std$(z) \leq 1.15$, $\max|z| \approx 2.1$, KS $p \in [0.37, 0.79]$, mean 1σ posterior widths 0.15–0.36 dex); **J-factor recovery was unbiased to ~ 0.1 dex** with std$(z) \leq 1.18$, $\max|z| \approx 2.3$, and a small high-side median bias of $+0.08$ to $+0.12$ dex. The Asimov decomposition below is re-verified under Jeffreys and matches the historical loguniform numbers to within ~ 0.01 dex.

5.8 Asimov-bias decomposition

The Asimov chain (1803 equal-weight samples; `tests/integration/analyze_asimov.py`) isolates the origin of the J-factor median offset.

(a) MAP vs. median vs. truth. Pushing through $J(\alpha_c)$:

Estimator	Offset from truth (dex)
1D-KDE MAP of $\log_{10} J(\alpha_c)$	+0.037
Chain median of $\log_{10} J(\alpha_c)$	+0.162
Chain mean of $\log_{10} J(\alpha_c)$	+0.214

The posterior peak (MAP) sits within 0.02–0.13 dex of truth at all four angles; the offset lives entirely in the median of a positively skewed transformation distribution.

(b) Small-x analytic check. In the small-x NFW limit, $J \propto \rho_s^2 r_s^3$. The chain median of $2\log_{10}\rho_s + 3\log_{10} r_s$ is +0.42 dex, against an actual $J(0.1°)$ bias of +0.13 dex. Per-sample regression of $\Delta\log_{10} J$ on the small-x prediction gives slopes 0.29–0.53 (not unity) and $r^2 = 0.08$–0.52 across the four angles — the log-linear small-x attribution does not hold over the full chain support.

(c) 1D-projection decomposition at α_c.

Projection	Median bias (dex)
$\log r_s$ only ($\log\rho_s$ = truth)	+0.45
$\log\rho_s$ only ($\log r_s$ = truth)	−0.32
Sum	+0.13
Full joint chain	+0.16
Non-additive remainder	+0.03

The full bias is a small residual between two large opposite-sign 1D-projection offsets. The non-additivity is negligible (0.02/0.16 dex), indicating that the chain ridge is approximately log-linearly aligned with the J surface. The J/D bias ratio across angles is 2.1–2.8, consistent with ρ^2 vs. linear ρ kernels both experiencing the same posterior-curvature skew. The pipeline publishes median + q_{16}/q_{84} + 1D-KDE MAP for every $\log_{10} J(\theta)$ and $\log_{10} D(\theta)$, so the median–MAP gap is visible to downstream consumers without any correction applied.

6 Tests

6.1 Unit tests

Unit tests reside in `tests/unit/` and are run with PYTEST. Each file targets one module or contract.

`tests/unit/test_jeans.py` verifies the Jeans solver across four categories: (1) the NFW auxiliary $g(x)$ via the small-x Taylor series and the stable-regime $\ln(1+x)$ form; (2) a $\tilde{\beta}$ prior-edge smoke test producing finite, positive $\sigma_{\rm los}$ with zero `RuntimeWarning`s across $\tilde{\beta} \in \{-0.95, -0.5, 0, +0.5, +0.95\}$, including the tangential extreme $\beta \approx -38$; (3) physical sanity — doubling ρ_s multiplies $\sigma_{\rm los}$ by $\sqrt{2}$ to 10^{-3} and $\sigma_{\rm los}(r_p)$ falls within a factor of three of the Wolf et al. (2010) estimator; (4) a 100-star likelihood call completes in under 50 ms on a single core.

`tests/unit/test_jeans_vs_quad.py` cross-checks the production log-grid integrator against `scipy.integrate.quad` (treated as ground truth, $\sim 10^4\times$ slower). Inner Jeans integrands and line-of-sight projections are compared across eight parameter combinations spanning the UFD and classical regimes at isotropic, radial, and tangential β. Tolerances 10^{-3} (inner) / 10^{-2} (projection). Run on demand before merging changes to integration tolerances or grid sizes.

`tests/unit/test_multi_epoch.py` verifies the multi-epoch primitives. IVW combination is checked against the textbook formula and against a closed-form expectation for $\sigma_{\rm sys} > 0$ added in quadrature post-combine. The χ^2 test is checked at zero scatter (no flag), at a 5σ outlier (flag), and for single-epoch input ($\chi^2, p =$ NaN, no flag). The strict-deconvolution path is verified to equal the as-statistical path when $\sigma_{\rm sys} = 0$, to return a strictly larger $\sigma_{\bar{v}}$ at fixed published total error, to raise `ValueError` when the implied $\sigma_{\rm stat}$ would be imaginary, and (single-epoch) to return $\sigma_{\bar{v}} = \sigma_{\rm total}$ exactly. Strict χ^2 uses $\sigma_{\rm stat}$ rather than $\sigma_{\rm total}$, producing a larger statistic.

`tests/unit/test_default_combiner.py` verifies the zero-point offset hook. Empty offsets reproduce the no-shift IVW. With offsets, velocities are shifted additively before IVW and the output recovers the common-zero-point $\bar{v}$. The combiner raises `KeyError` on missing `Inst` column, raises on unknown instrument tag, and does not mutate the caller's input array.

`tests/unit/test_combiner_dispatch.py` checks the per-paper combiner registry. Each of the six per-epoch bibcodes (Li+2017/2018, Chiti+2022/2023, Simon+2020, Hansen+2024) dispatches to a dedicated handler. An unknown bibcode falls back to the default combiner with $\sigma_{\rm sys} = 0$ and $p_{\rm threshold} = 0.01$.

`tests/unit/test_preprocess.py` exercises `prepare_jeans_input`. Per-star catalogs bypass the combiner; per-epoch catalogs invoke the registered handler followed by selection. A two-star, two-epoch example verifies that a variable star is dropped by the variability cut. Policies are threaded through and appear in the audit dict. Missing `source_paper_bibcode` on a per-epoch catalog and unknown `catalog_granularity` both raise `ValueError`. A parametrised smoke test runs every per-epoch catalog through the full pipeline.

`tests/unit/test_priors.py` checks the prior registry. Exactly the three keys `uniform`, `loguniform`, `jeffreys` are registered; an unknown key raises. The log-uniform transform is checked byte-for-byte over 1000 unit-cube draws. The Jeffreys log-correction returns $-\infty$ when all T_i are equal (degenerate Fisher determinant) and a finite value otherwise; the `needs_T` flag is checked, and the correction returns exactly zero for the uniform and log-uniform priors.

`tests/unit/test_selection.py` verifies `select_jeans_stars`. Membership, radial, and variability cuts are exercised in isolation and in combination, including a regression for a bug where uniform $p \in \{0, 1\}$ membership was mistaken for a pre-applied cut. Disabling the variability cut retains all stars. Missing `R`, invalid $r_{1/2}$, and per-epoch input each raise. An `NpzFile` compatibility test verifies that `np.load` output is accepted directly.

6.2 Integration tests

6.2.1 Segue 1 end-to-end test

`tests/integration/run_segue1.py` runs the complete pipeline on Segue 1 and compares the output to published results. Full toggles, results history, and the P&S 2018 comparison table live in `docs/plan/segue1_test.md`; this section summarizes the test spec. Three per-star data sources are toggled by the `SOURCE` variable: the Simon et al. (2011) VizieR catalog with $B_{\rm pr} > 0.8$ (`simon`); the Pace pre-combined Bayes file with $p \geq 0.8$ (`pace`); and the Geha et al. (2026) catalog staged through the production `prepare_jeans_input` pipeline with $p_{\rm mem,novar} = 1$ and the $R < 2\, r_{1/2,\rm maj} \sqrt{1-\varepsilon}\,(4/3)$ sphericalised-3D-Plummer cut (`geha`); the variability removal is already folded into $p_{\rm mem,novar}$ upstream in Paper I. For `simon / pace`, nuisance priors follow P&S 2018: $d = 23 \pm 2\,$kpc, $r_{1/2,\rm 2D} = 21 \pm 5\,$pc ($r_h = 4.31 \pm 1.03\,$arcmin with $\varepsilon = 0.47 \pm 0.11$). For `geha`, nuisance priors are taken from LVDB v1.0.5. The reference is $\sigma_{\rm los} = 3.1^{+0.9}_{-0.8}\,{\rm km\,s^{-1}}$ from P&S 2018 Table A1. All three paths run 7-D dynesty inference and produce posterior samples, a summary CSV, and diagnostic plots in `results/tests/segue1/`.

6.2.2 UFD population MC recovery

`tests/integration/run_ufd_population.py` performs the 15-realisation MC of Section 4.4. All realisations share the same injected truth; realisations differ only in the random seed for stellar positions and velocity scatter. Each realisation runs dynesty (`nlive`=300, `dlogz`=0.5, log-uniform prior) or reloads a cached chain from `results/tests/ufd_population/`. Population diagnostics for five parameters ($\bar{V}$, $\log_{10} r_s$, $\log_{10}\rho_s$, $\tilde{\beta}$, $\log_{10}(\rho_s r_s^3)$) are written to `results/tests/ufd_population/ufd_pop_diagnostics.json`. A single Asimov realisation is run via the `--asimov` flag.

6.2.3 Asimov bias decomposition

`tests/integration/analyze_asimov.py` loads the chain produced by `--asimov` and performs six diagnostics: (1) $M(r_{1/2,\rm 3D})$ recovery (median, mean, 1D-KDE peak vs. truth, plus skewness and excess kurtosis); (2) full-chain push through J and D at four angles each; (3) small-x analytic test on the $2\log_{10}\rho_s + 3\log_{10} r_s$ proxy; (4) per-sample $\Delta\log_{10} J$ regression on the small-x prediction (slope and R^2); (5) marginal-projection decomposition at α_c holding one channel at truth and pushing the other; (6) prior-edge fraction within $\{0.05, 0.10, 0.20\}$ of the $\log_{10} r_s$ prior boundaries. Outputs are saved to `results/tests/ufd_population/asimov_chain_diagnostics.npz`; the headline numbers in Section 5.8 come from this run.

7 Stage 2 systematic corrections

Two corrections to the Stage 2 per-star likelihood live in this section. The Kaplinghat–Strigari perspective-motion correction (§7.1) is implemented end-to-end, with PM uncertainty marginalised inside Stage 2. Velocity gradients (§7.2) remain an open issue.

7.1 Perspective-motion correction

For satellites with measured bulk proper motion the Kaplinghat–Strigari perspective correction (Walker et al. 2008, Appendix; Kaplinghat & Strigari 2008) adjusts each per-star $v_{\rm los}$ by the projection of the galaxy's plane-of-sky bulk velocity onto the star's line of sight, which differs from the line of sight to the galaxy centre for stars at non-zero angular separation. In the small-angle limit appropriate to our sample (fields of $\lesssim 1^\circ$ across),

$$\Delta v_{\rm persp}(\alpha,\delta) = A\,d\left(\mu_{\alpha*}\,\Delta\alpha\cos\delta_0 + \mu_\delta\,\Delta\delta\right), \tag{47}$$

with $A = 4.74047\ {\rm km\,s^{-1}}\,({\rm mas\,yr^{-1}\,kpc})^{-1}$, d in kpc, $\mu_{\alpha*} = \mu_\alpha\cos\delta$ in the Gaia convention, and $(\Delta\alpha\cos\delta_0, \Delta\delta)$ the tangent-plane offset in radians; this is the limit of K&S eq. 1 with the $-\frac{1}{2}v_{\rm sys}\rho^2$ contraction term (a subtractive correction for receding systemic velocities) dropped. The corrected per-star velocity entering the Gaussian likelihood is $v_{\rm corr} = v_{\rm los} - \Delta v_{\rm persp}$.

Across the 39-galaxy study sample (LVDB v1.0.5, Pace+ 2022 proper motions), peak post-selection per-star shifts span 0.04–$3.72\,{\rm km\,s^{-1}}$ (median $\sim$$0.38\,{\rm km\,s^{-1}}$), with Leo V the low end and Boötes III the high end. Galaxies whose spectroscopic catalogues extend many half-light radii beyond the analysis cut (Carina III: $\rho_{\rm cat}/R_h \simeq 12.7$) show much larger raw shifts on the parent catalogue ($\sim$6 km s^{-1} for Carina III) that collapse to $<$0.2 km s^{-1} on the actual Stage 2 sample. The post-selection magnitudes remain comparable to or above the per-star velocity uncertainties for the wider-field galaxies and the high-proper-motion ultra-faints, and motivate applying the correction throughout the sample. The dropped quadratic $\frac{1}{2}v_{\rm sys}\rho^2$ term remains below 0.05 km s^{-1} at $\rho < 0.5°$ for $v_{\rm sys} \lesssim 300$ km s^{-1}, justifying the small-angle approximation; the helper `perspective_correction_full` retains the full K&S expression as a cross-check.[1] Ingest now exposes $\mu_{\alpha*}$, μ_δ and their asymmetric errors as registry columns (`pmra_mas_yr`, `pmdec_mas_yr`, `pmra_em_mas_yr`, ...) and propagates them to the per-galaxy `_meta`; the compute module is `dwarfjeans.jeans.perspective`. `prepare_jeans_input` subtracts $\Delta v_{\rm persp}$ at the LVDB-published $(\mu_{\alpha*}, \mu_\delta)$ after the Stage 2 selection cut, stashes the observed $v_{\rm los}$ as `V_observed`, and emits the diagnostics into `audit['perspective']`. Stage 2 then marginalises over the proper-motion uncertainty: `make_loglike_with_nuisances` extends the nuisance vector from 7D to 9D by sampling $(\mu_{\alpha*,\rm draw}, \mu_{\delta,\rm draw})$ from split-normal priors built from the LVDB asymmetric errors, and recomputes $\Delta v_{\rm persp}$ per draw using the *sampled* distance d so that the distance prior propagates correctly through both the geometric and perspective terms.

A mock-data calibration on an Antlia-II-like configuration ($N_\star = 50$, $d = 124$ kpc, $R_h \approx 1.4°$, $|\mu| \approx 0.14$ mas yr^{-1}; $N = 8$ realisations at `nlive`= 100) detects no $\sigma_{\rm los}(R_h)$ bias from the perspective + PM pipeline at the standard-error sensitivity the run achieves: median-of-medians = 12.246 km s^{-1} vs truth 12.232 km s^{-1}, SE $\approx$ 0.51 km s^{-1} ($\approx$ 4% of truth) from the realisation-to-realisation dispersion of 1.45 km s^{-1}. This isolates the correction itself: nuisance priors are pinned at truth and the PM truth equals the prior central, so this run tests the perspective + PM *plumbing*, not the full production end-to-end recovery under realistic nuisance priors or under a data-vs-prior PM disagreement. A wider sweep (more realisations, offset PM truth, production-realistic nuisance priors) is left as a follow-up before any tightened precision claim.

7.2 Velocity gradients

The Stage 2 likelihood currently assumes a single zero-mean (systemic-subtracted) dispersion with no spatial trend in $\langle v_{\rm los}\rangle$. Several classical dwarfs (e.g. Carina, Fornax, Sculptor) show measurable line-of-sight gradients from a mix of intrinsic rotation, tidal streaming, and the perspective-motion signal of §7.1. With the perspective correction now in place, the residual gradient is the rotation/streaming component and the two contributions must not be double-counted. Three options are under consideration — ingest-time subtraction of a measured per-galaxy gradient, in-likelihood marginalisation over an amplitude–position-angle pair, or documenting that the ultra-faint gradients are consistent with zero and skipping the correction in that regime. The implementation is deferred and tracked alongside the perspective correction.

[1] Antlia II is the one exception worth flagging: with $R_h \simeq 1.7°$ the post-selection sample reaches $\rho_{\rm max} \simeq 2.2°$, where the dropped $\frac{1}{2}v_{\rm sys}\rho^2$ term reaches $\sim$0.21 km s^{-1} — about 7% of the linear shift's peak (2.99 km s^{-1}) and the largest such fraction in the sample. The correction itself shifts $\sigma_{\rm los}$ by +0.65 km s^{-1} (+10%, $\sim 2\sigma_{\rm stat}$); independent verification confirms the sign — for Antlia II the per-star perspective shift is anti-correlated with the velocity residual ($r = -0.43$), so the gradient was partially cancelling the intrinsic dispersion in $v_{\rm los}$ and removing it broadens the distribution rather than narrowing it. A spatial-jackknife test to separate the genuine bulk-PM gradient from catalog-sampling effects, and a switch to `perspective_correction_full`, are left for any future tightening of precision claims on Antlia II.

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[TODO: Surveys, software, funding.]